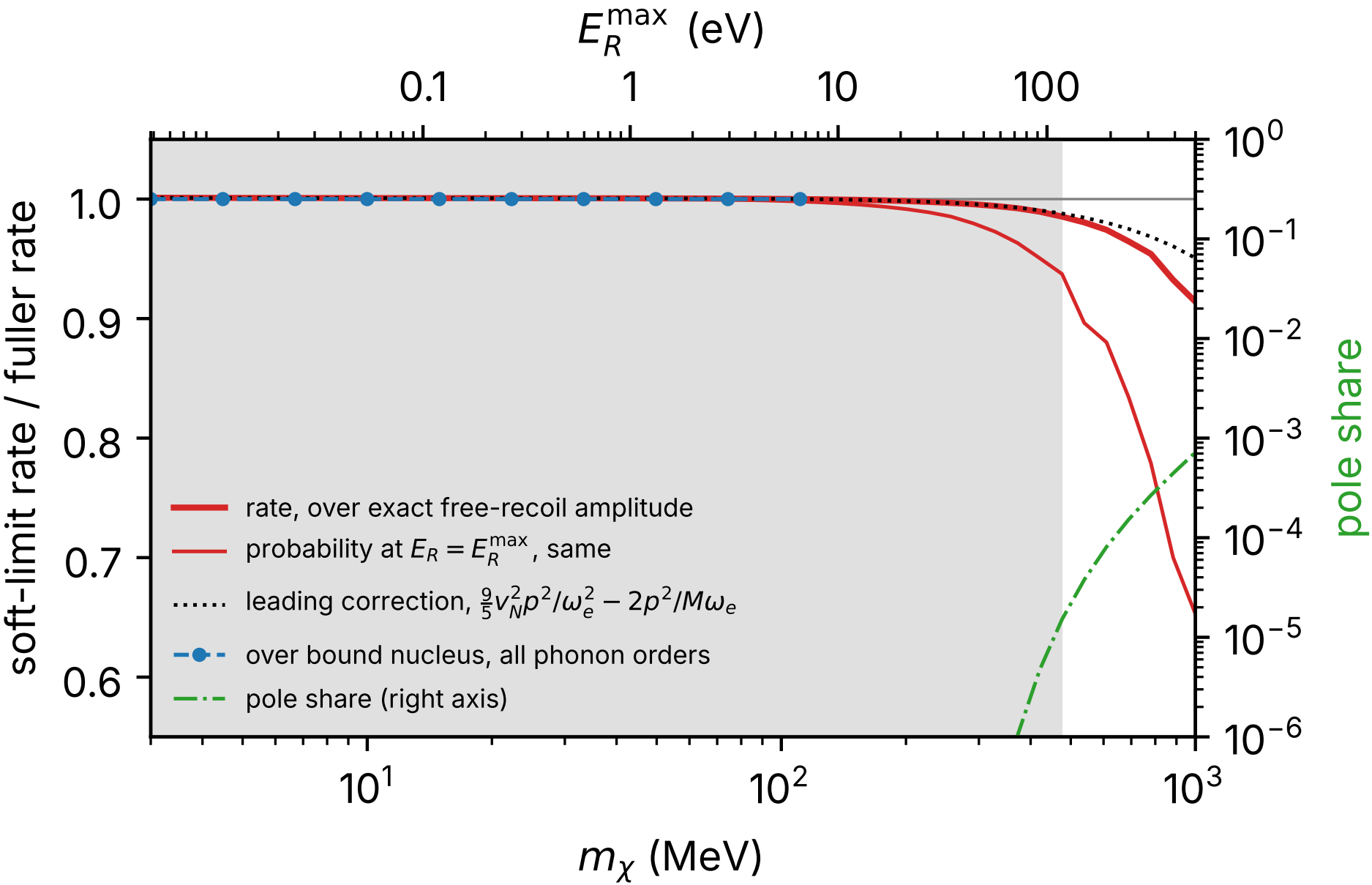
$E_R^{\max}$ (eV)
0.1
1
10
100
soft-limit rate / fuller rate
1.0
0.9
0.8
0.7
0.6
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
pole share
rate, over exact free-recoil amplitude
probability at $E_R = E_R^{\max}$, same
leading correction, $\frac{9}{5}v_N^2 p^2/\omega_e^2 - 2p^2/M\omega_e$
over bound nucleus, all phonon orders
pole share (right axis)
10^1
10^2
10^3
m_χ (MeV)